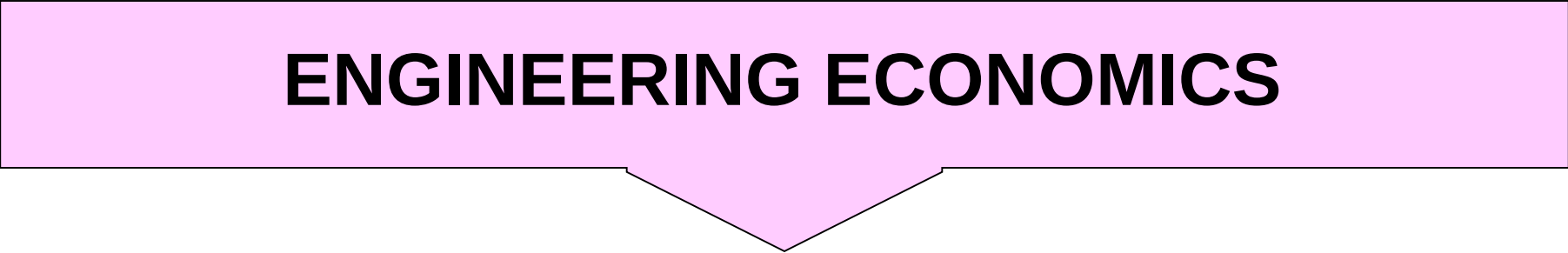


ENGINEERING ECONOMICS



LECTURE - 10



ECONOMIC EQUIVALENCE

- Alternatives should be compared as far as possible when they produce similar results, serve the same purpose or accomplish the same function.
- How can alternatives for providing the same service or accomplishing the same function be compared when interest is involved over extended periods of time?

➤ Consider the comparison of alternative options, or proposals, by reducing them to an equivalent basis, depending on:

1. interest rate
2. amounts of money involved
3. timing of the monetary receipts and/or expenditures
4. manner in which the interest , or profit on invested capital is paid and the initial capital is recovered.

- To better understand the concept of economic equivalence, consider a situation in which we borrow \$8000 and agree to repay it in four years at an interest rate of 10% per year. There are many plans by which the principal of this loan and the interest on it can be repaid.
- For simplicity, consider four plans, in each plan the interest rate is 10% per year and the original amount borrowed is \$8000. Thus the difference among the plans rest with items (3) and (4).

Note: If two alternatives are economically equivalent, then they are equally desirable to the borrower.

Comparison of Plans

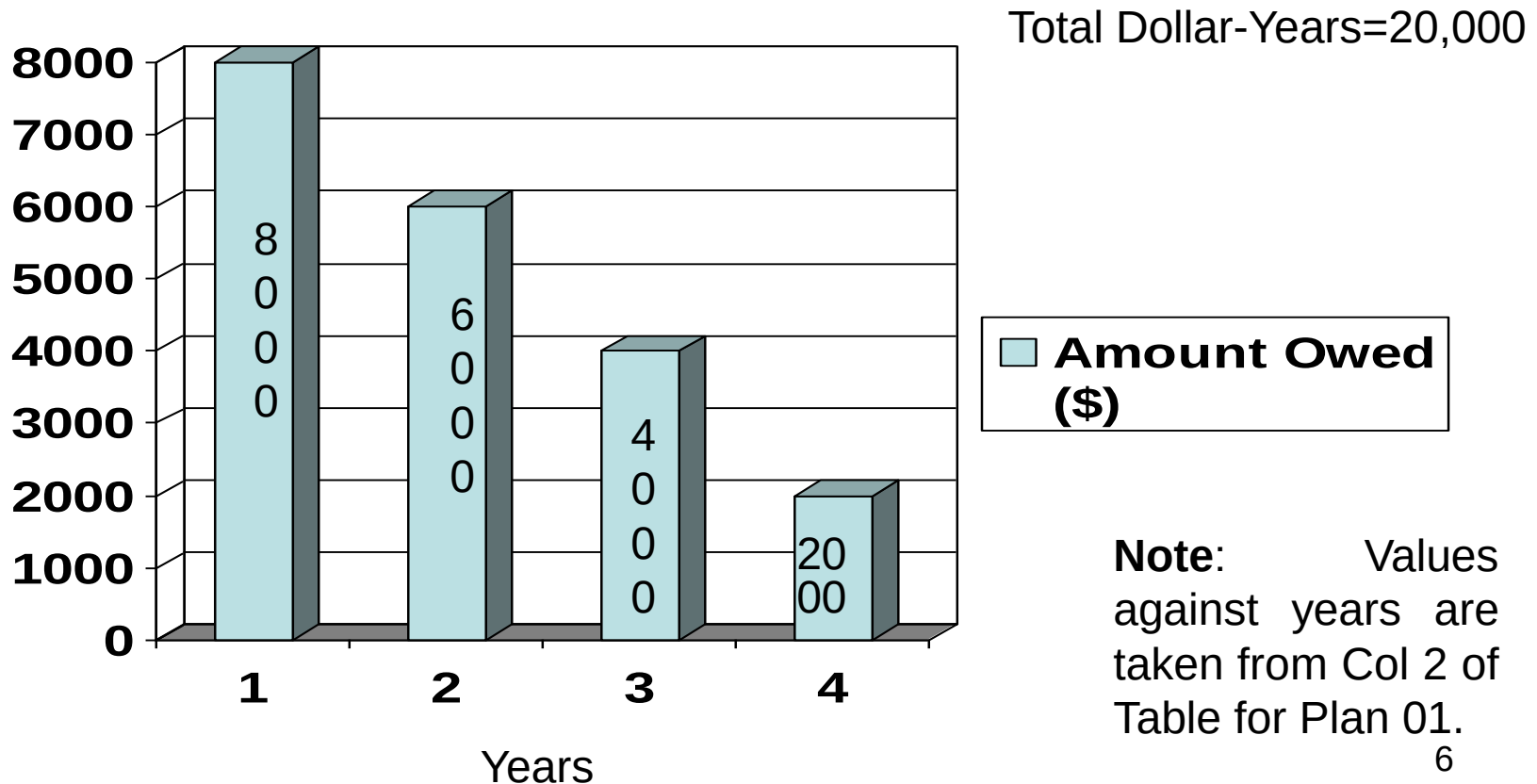
The four plans are shown on next slides and it will soon be apparent that all are equivalent at an interest rate of 10% per year.

Plan 04 involves compound interest. The total amount of interest repaid in plan 04 is highest of all the plans considered.

Economic equivalence is established, in general, when we have indifference between a future payment or series of future payments, and a present sum of money.

To see why the four plans are equivalent at 10%, we could plot the amount owed at the beginning of each year (column 02) versus the year.

The area under the resulted bar chart represents the dollar years that the money is owed. For example, the dollar years for plan 01 equals 20,000, which is obtained from this graph.



Comparison of Plans 1 - 4 (see pp. 71-73)

Summing the amount owed at the beginning of each year yields the amount of money "rented" in dollar-years.

Plan	Area under curve (Dollar-years)	Total Interest Paid	Ratio of Total Interest to Dollar-Years
1	\$20,000	\$2,000	0.10
2	32,000	3,200	0.10
3	20,960	2,096	0.10
4	37,130	3,713	0.10

Equivalence is established when total interest paid, divided by dollar-years of borrowing, is a constant ratio among financing plans.

- Because the ratio is constant at 0.10 for all plans, we can deduce that all repayment methods considered are equivalent, even though each involves a different total end of year payment.
- In summary, equivalence is established when total interest paid, divided by dollar-years of borrowing, is a constant ratio among financing plans.

Conclusion

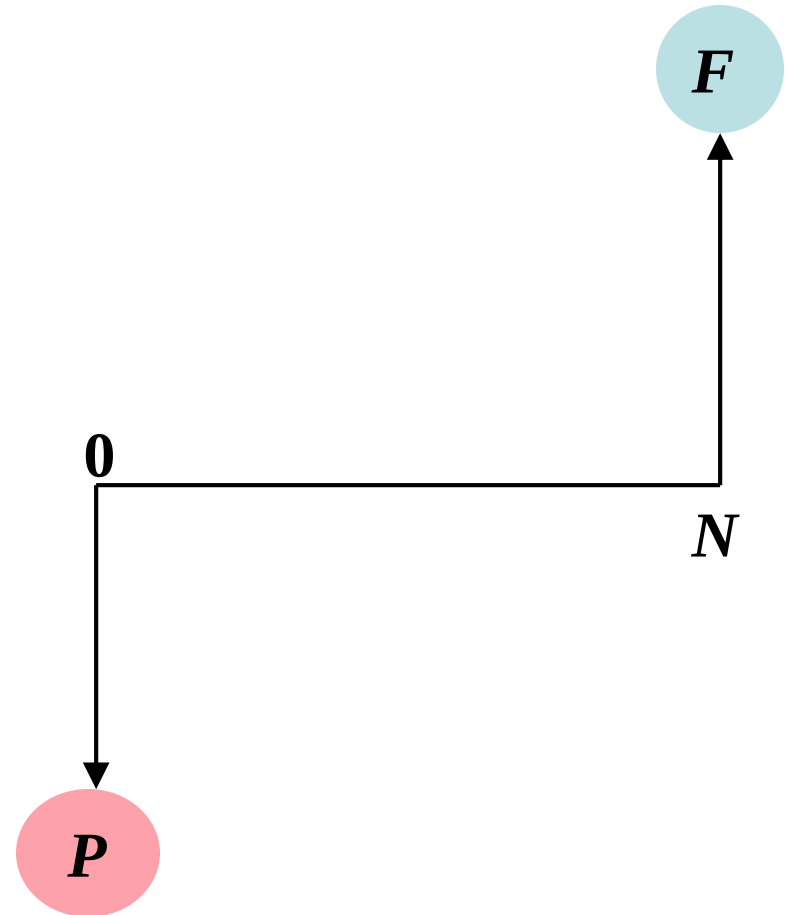
- **Economic equivalence** exists between cash flows that have the same economic effect and could therefore be traded for one another.
- Even though the amounts and timing of the cash flows may differ, the **appropriate interest rate** makes them equal.

Equivalence from Personal Financing Point of View

- If you deposit P dollars today for N periods at i , you will have F dollars at the end of period N .

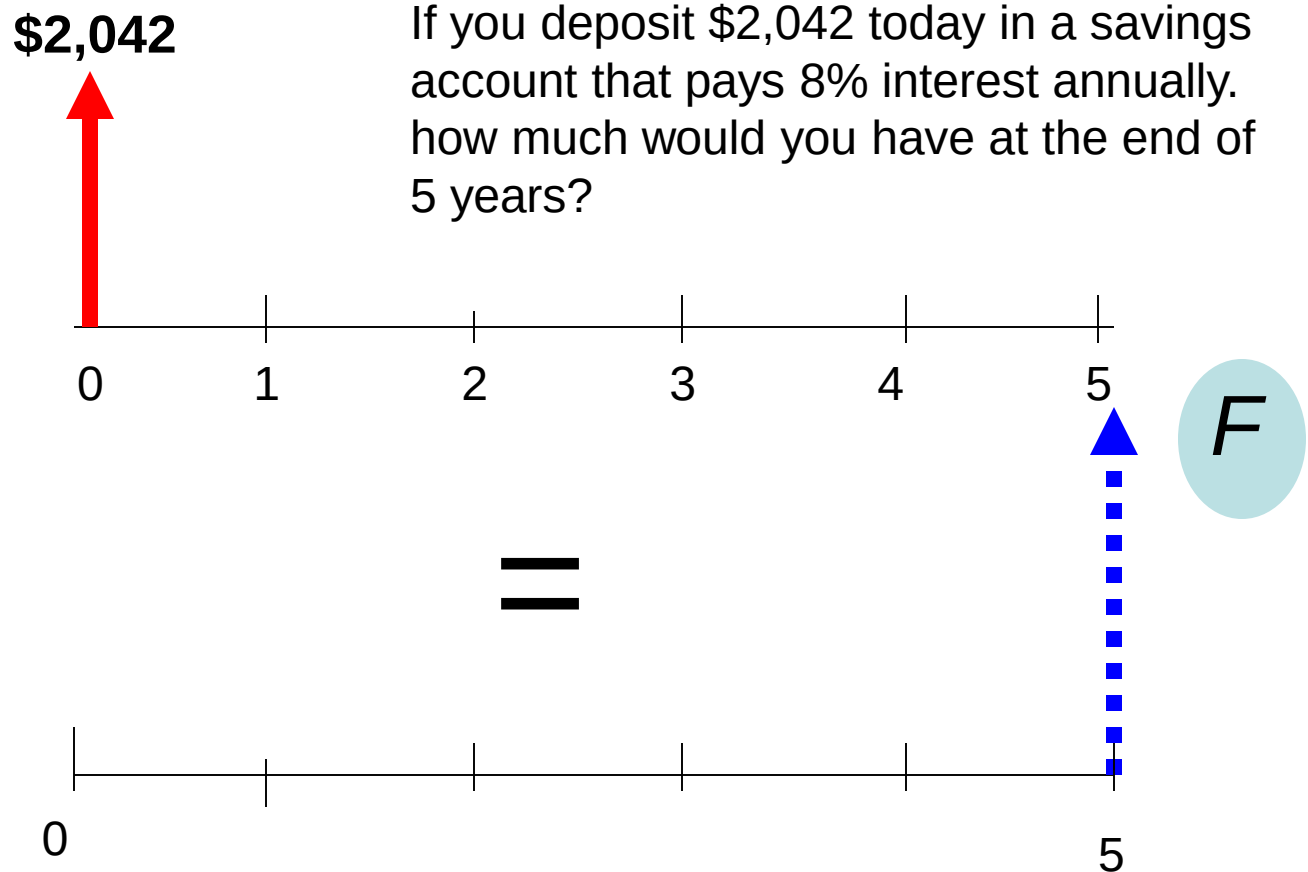
$$F = P(1+i)^N$$

Note: This formula will calculate compound interest.



Practice Problem 01

At 8% interest, what is the equivalent worth of \$2,042, 5 years from now?

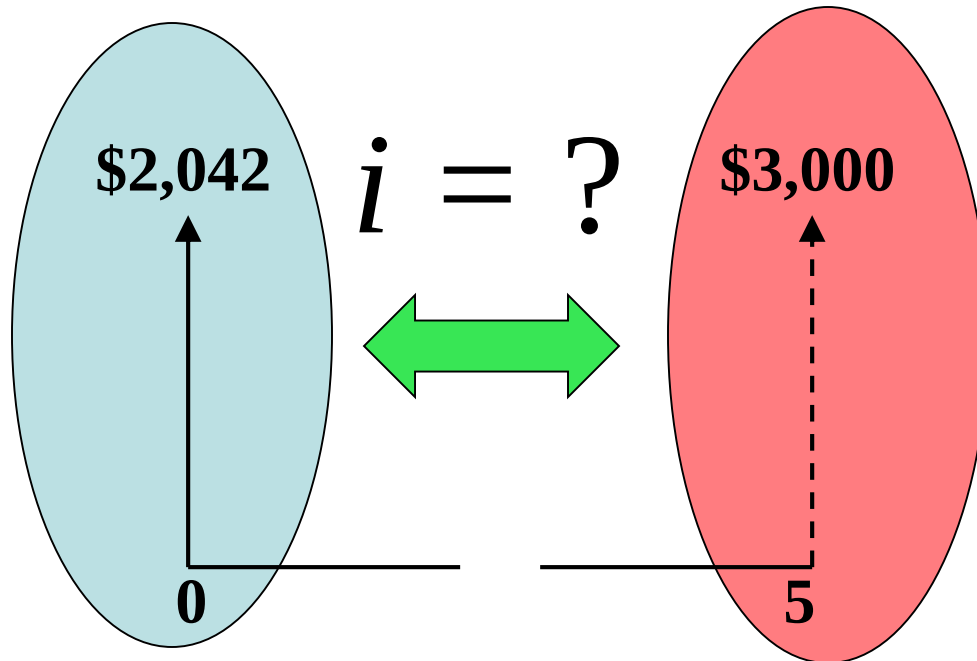


Solution 01

$$\begin{aligned} F &= \$2,042(1 + 0.08)^5 \\ &= \$3,000 \end{aligned}$$

Practice Problem 02

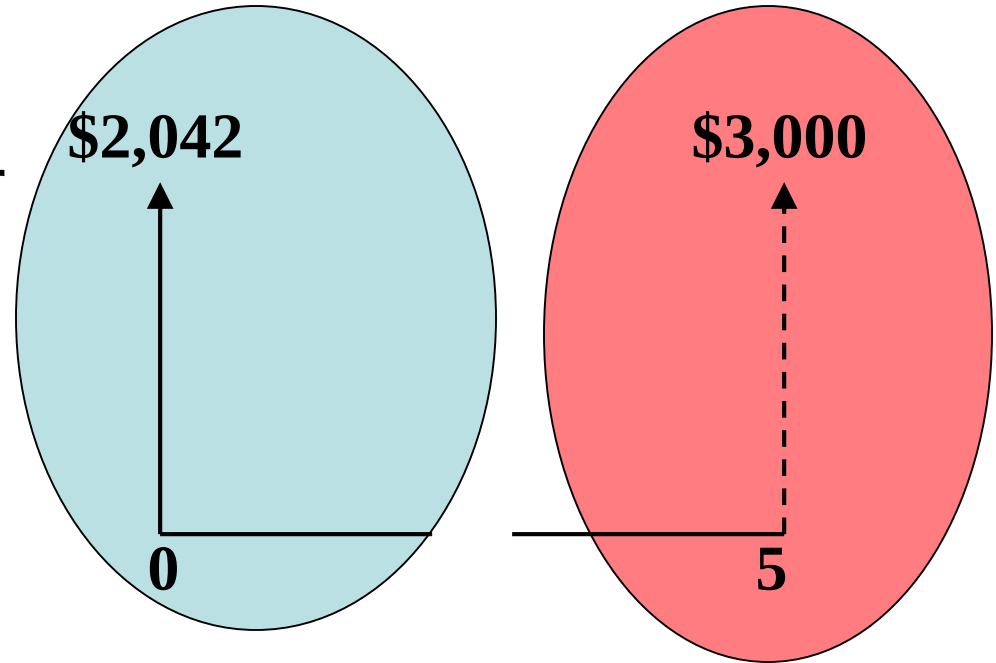
At what interest rate
would these two amounts be equivalent?



Solution 02

Equivalence Between Two Cash Flows

- **Step 1:** Determine the base period, say, year 5.
- **Step 2:** Identify the interest rate to use.
- **Step 3:** Calculate equivalence value.



$$i = 6\%, F = \$2,042(1 + 0.06)^5 = \$2,733$$

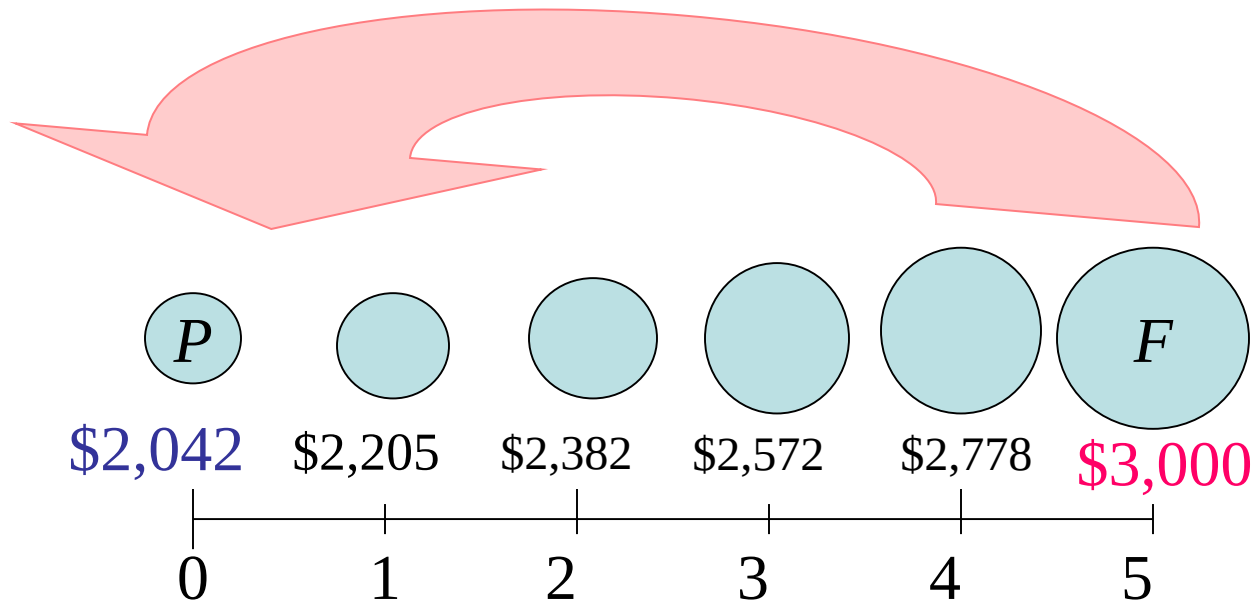
$$i = 8\%, F = \$2,042(1 + 0.08)^5 = \$3,000$$

$$i = 10\%, F = \$2,042(1 + 0.10)^5 = \$3,289$$

Example - Equivalence

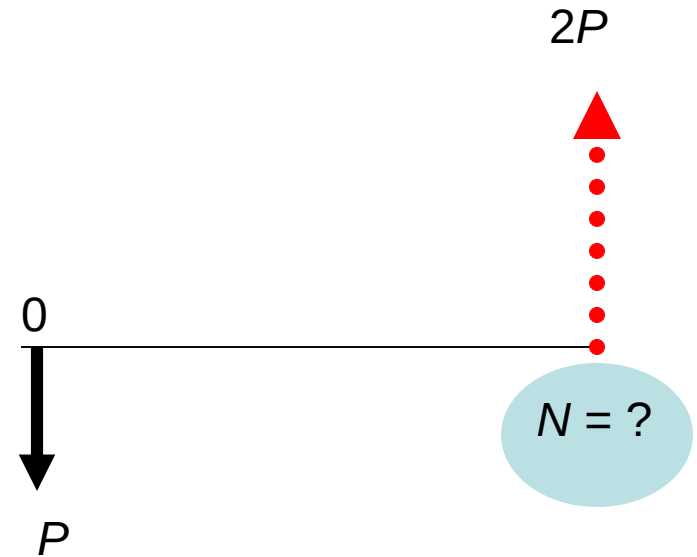
Various dollar amounts that will be economically equivalent to \$3,000 in 5 years, given an interest rate of 8%.

$$P = \frac{\$3,000}{(1+0.08)^5} = \$2,042$$

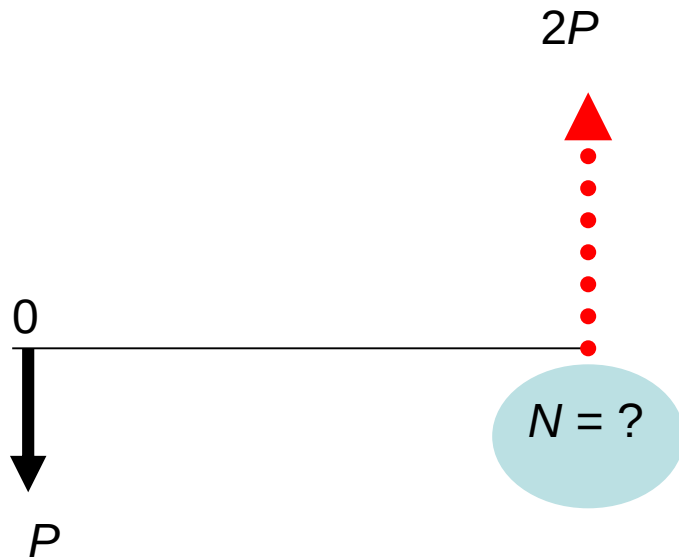


Practice Problem 03

- How many years would it take an investment to double at 10% annual interest?



Solution 03



$$F = 2P = P(1 + 0.10)^N$$

$$2 = 1.1^N$$

$$\log 2 = N \log 1.1$$

$$N = \frac{\log 2}{\log 1.1}$$

$$= 7.27 \text{ years}$$

Rule of 72

Approximating
how long it will
take for a sum of
money to double

$$\begin{aligned} N &\cong \frac{72}{\text{interest rate (\%)}} \\ &= \frac{72}{10} \\ &= 7.2 \text{ years} \end{aligned}$$

Important Note:

For more detail, read topics 3.1 to 3.9 from Engineering Economy (eleven edition) by William G Sullivan along with problems.

Discussion